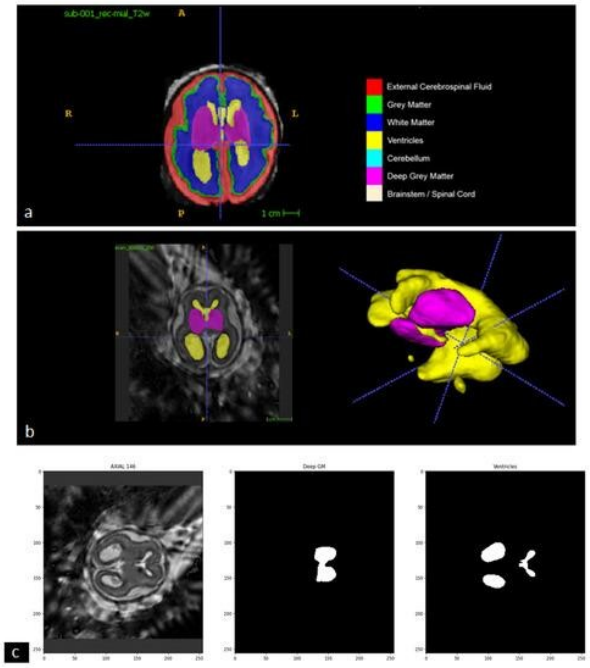
Project 1: Explainable Deep Learning for Fetal MRI Classification



Objective

Develop a deep-learning model to classify fetal brain MRI scans as neurotypical (normal) or affected by spina bifida, using the FeTA dataset. Investigate whether an attention mechanism can highlight the brain regions that contribute most to the model's predictions.

The project combines medical image classification with explainable AI: students should evaluate not only whether the model classifies scans correctly, but also where it looks when making a prediction.

Proposed tasks

1. Dataset preparation: obtain the refined FeTA annotations, identify neurotypical and spina bifida cases, and exclude other pathologies. Record gestational age and whether spina bifida cases were scanned before or after surgery.
2. Preprocessing: normalize 3D T2-weighted MRI volumes, standardize spatial dimensions, and optionally use the provided tissue segmentation maps.
3. Baseline classification: train a 3D CNN to distinguish neurotypical from spina bifida scans. Compare it with a simpler model using extracted MRI features.
4. Attention and explainability: add an attention module or use 3D Grad-CAM to generate heatmaps showing which brain regions contribute to predictions. Overlay the heatmaps on the MRI scans.
5. Evaluation: report sensitivity, specificity, ROC-AUC, F1-score and confusion matrices. Compare attention maps across correctly and incorrectly classified cases.
6. Discussion: assess whether the model identifies plausible anatomical patterns or exploits confounders such as gestational age, acquisition site or surgery status.

Expected outcome

A reproducible classification pipeline, quantitative performance comparison, and MRI visualizations showing the regions associated with the model's decisions.

Important feasibility check: the standard FeTA release provides neurotypical/pathological labels, but not necessarily the diagnosis for every case. The refined dataset paper reports the spina bifida diagnoses, so students must first verify access to the corresponding subject-level diagnosis labels and MRI files. The reported 36 spina bifida cases include pre- and post-surgical cases; these should be tracked separately. With such a small cohort, use subject-level splits and preferably cross-validation, and treat the results as exploratory rather than clinically validated.

Dataset and reference papers

1. FeTA fetal MRI dataset

Public fetal brain MRI reconstructions with tissue segmentation maps and neurotypical/pathological labels. The refined dataset reports 34 neurotypical and 54 pathological cases, including 36 spina bifida cases (pre- and post-surgery). Use only those 36, unless you have cool ideas
Official FeTA dataset and download instructions

Refined dataset paper and diagnosis information

2. Hosna Asma ull et al. , developmental brain MRI review

Developmental Brain Age Estimation From MRI Data: A Systematic Review of Deep Learning Approaches and Open Datasets.

Use this review to explore fetal and developmental MRI datasets, existing deep-learning approaches, and the limitations of available data.

Read the review

3. Subbarao et al. (2025) , ventriculomegaly classification

This paper uses FeTA scans, ventricular segmentation, automated measurements and deep-learning classification. It provides a methodological reference for building a fetal MRI classification pipeline. Read the open-access paper

Project 2 CEBRA for pre-seizure data visualization and prediction.

Objectives

The goal is to investigate whether dimensionality-reduction and representation-learning methods can reveal meaningful patterns in neonatal EEG and support seizure-state classification, compare:

- PCA: linear dimensionality reduction.
- UMAP and t-SNE: nonlinear embedding and visualization.
- CEBRA-Time: contrastive learning of temporal representations.

The project has two components:

1. Unsupervised analysis: Learn and visualize EEG embeddings without using seizure labels during training. Investigate whether normal, pre-ictal, and ictal segments exhibit distinct latent structures.
2. Supervised classification: Train a common downstream classifier on each representation to distinguish normal/interictal, pre-ictal, and ictal EEG.

Methodology

1. **Preprocess EEG** and extract fixed-length windows.
2. Learn embeddings using PCA, UMAP, t-SNE, and CEBRA-Time.
3. Visualize latent spaces and investigate their temporal structure.
4. Train comparable classifiers on each representation.
5. Evaluate using subject-independent splits, ensuring that recordings from the same infant never appear in both training and test sets.
6. Compare performance, embedding stability, and generalization.

Define pre-ictal windows using a prespecified interval before seizure onset. Pre-ictal classification does not by itself demonstrate seizure prediction.

Evaluation

Report balanced accuracy, macro-F1, AUROC, AUPRC, and per-class sensitivity. Where feasible, evaluate seizure-level detection, false alarms per hour, and warning time.

Research question

Does temporal contrastive learning with CEBRA provide more useful and generalizable EEG representations than PCA, UMAP, and t-SNE for neonatal seizure-state classification?

Expected outcome: A reproducible pipeline, embedding visualizations, a classification benchmark, and a critical comparison of the methods.

Resources

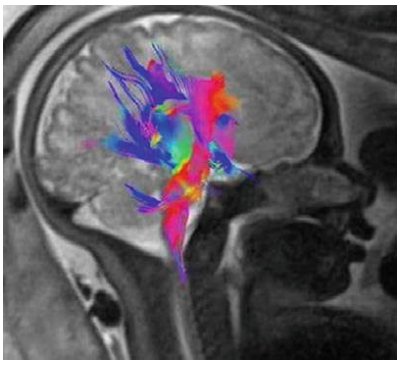
Datasets

- Primary , Helsinki neonatal seizure EEG: https://zenodo.org/records/4940267 Multichannel neonatal EEG with expert seizure annotations.
- External validation , Siena Scalp EEG: https://physionet.org/content/siena-scalp-eeg/1.0.0/ An independent epilepsy dataset from an older population, suitable for exploratory cross-population validation but not direct neonatal validation.

  Look for other epilepsy EEG dataset
- CEBRA repository: https://github.com/AdaptiveMotorControlLab/CEBRA
- EEG processing: https://mne.tools/

Project 3: Explainable Graph Attention Networks

for Preterm Birth Classification Using Neonatal Diffusion MRI



Background. Premature birth can affect the development of cerebral white matter and the organization of structural brain connectivity. Diffusion MRI (dMRI) provides information about white-matter microstructure, while tractography can be used to reconstruct structural connections between brain regions. Although previous studies have investigated preterm-versus-term differences in neonatal structural networks, graph attention networks (GATs) and explainable AI offer opportunities to investigate which brain regions and connections are most informative for classification.

Objective. The aim of this project is to develop an explainable deep-learning framework to distinguish preterm-born infants from term-born controls using neonatal diffusion MRI from the Developing Human Connectome Project (dHCP). The project will investigate whether graph attention networks can identify structural connectivity patterns associated with prematurity and determine which brain regions, white-matter connections, or individual tracts contribute most strongly to the model's predictions.

Methodology

1. **Dataset preparation.** Obtain neonatal diffusion MRI and associated demographic and clinical metadata from dHCP. Define preterm-born infants and term-born controls using gestational age at birth. Record postmenstrual age at scanning and other relevant covariates, and apply appropriate imaging quality-control procedures.
2. **Diffusion processing and feature extraction.** Process the diffusion-weighted images to estimate diffusion tensor metrics, including fractional anisotropy (FA), mean diffusivity (MD), axial diffusivity (AD), and radial diffusivity (RD). Reconstruct white-matter pathways using tractography and derive connectivity matrices and tract-specific measurements for major white-matter bundles where reliably identifiable.
3. **Graph construction.** Represent each infant's structural connectome as a graph, with brain regions as nodes and tractography-derived connections as edges. Node features may include regional diffusion measurements, while edge features may include connection strength, streamline counts, and tract-level diffusion statistics. Compare alternative graph representations, including region-level structural connectomes and tract-specific feature graphs.
4. **Classification models.** Establish conventional machine-learning baselines using diffusion and tract-level features, followed by a graph convolutional network and a graph attention network (GAT). Evaluate whether attention-based message passing improves classification performance and supports more interpretable predictions.

5. **Explainable AI.** Investigate model explanations using attention weights, graph explainers, and feature-attribution methods. Identify the brain regions, connections, and white-matter tracts most relevant to preterm-versus-term classification. Visualize the resulting explanations on anatomical brain representations and compare the consistency of important features across cross-validation folds.
6. **Evaluation and validation.** Evaluate the models using subject-level train/test splits or nested cross-validation, reporting ROC-AUC, balanced accuracy, sensitivity, specificity, and F1-score. Control for potential confounding factors, particularly postmenstrual age at scanning, gestational age at birth, sex, and acquisition-related differences where applicable. Compare the performance of graph-based models with conventional machine learning and diffusion-image-based baselines.

Expected outcomes

The project is expected to deliver a reproducible neonatal diffusion MRI processing and classification pipeline, a comparison of conventional and graph-attention models, and anatomical visualizations of the regions and structural connections associated with model predictions. The results may help characterize which aspects of neonatal white-matter microstructure and structural connectivity are most informative for distinguishing preterm-born infants from term-born controls.

Dataset and references

- **Dataset:** Developing Human Connectome Project (dHCP)
- **Neonatal diffusion MRI processing pipeline:** Bastiani et al., 2019 , Automated processing pipeline for neonatal diffusion MRI in the developing Human Connectome Project
- **Related structural-network classification study:** Preterm-birth alters the development of nodal clustering and neural connection pattern in brain structural network at term-equivalent age
- **Related structural-connectome study:** Network controllability of structural connectomes in the neonatal brain

Project 4: Cross-Modal Brain Registration from MRI to Polarized Light Imaging (PLI)

Project description

Objective. Develop and evaluate a registration pipeline to align MRI brain images with higher-resolution Polarized Light Imaging (PLI) data. The aim is to transfer anatomical information from MRI to PLI while accounting for differences in spatial resolution, contrast, and tissue appearance.

The project will use data from the Oxford Digital Brain Bank, which provides access to human brain imaging resources. Students should first investigate which subjects have both MRI and PLI data available and determine the appropriate registration targets.

Tasks

1. **Dataset exploration:** identify subjects with corresponding MRI and PLI data. Inspect image dimensions, voxel spacing, anatomical coverage, and available metadata. Determine whether the images are already approximately aligned or require substantial registration.
2. **Preprocessing and subsampling:** preprocess the MRI and PLI data and bring them into a common coordinate system where possible. Because MRI generally has much lower spatial resolution than PLI, investigate subsampling or aggregation of PLI data to an MRI-compatible resolution for registration and validation. Preserve the original high-resolution PLI data for the final transformation and visualization.
3. **Registration pipeline:** implement a suitable MRI-to-PLI registration method, starting with rigid and affine transformations and then exploring deformable registration if justified. Consider tools such as ANTs or SimpleITK. Since MRI and PLI have different image contrasts, evaluate cross-modal similarity measures rather than relying exclusively on raw intensity matching.
4. **Registration validation:** investigate how to assess alignment independently of the similarity measure optimized during registration. Possible approaches include:
    - Comparing anatomical landmarks or manually identified corresponding points before and after registration.
    - Measuring landmark target registration error, where suitable landmarks are available.
    - Comparing brain boundaries or tissue interfaces using distance-based metrics.
    - Checking alignment of identifiable anatomical structures across registered slices.
    - Performing sensitivity analyses to determine whether results change substantially with subsampling resolution, initialization, or registration settings.
5. **Visualization and analysis:** display MRI and PLI images side by side and as overlays, checkerboards, or contour plots. Document successful registrations, failure cases, and anatomical regions where alignment remains uncertain.

Expected outcomes

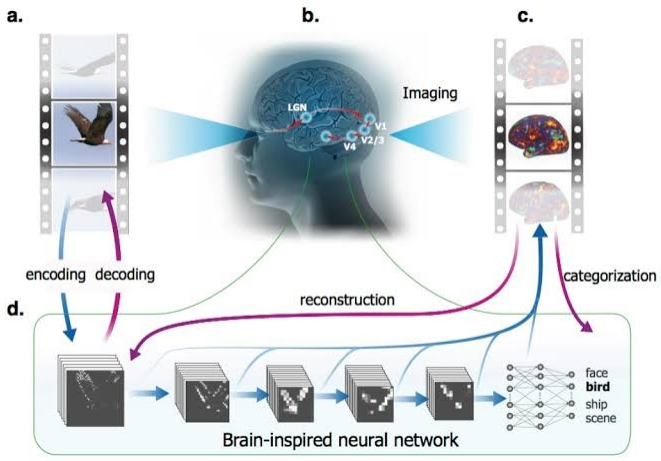
A reproducible cross-modal registration pipeline, quantitative and visual validation of registration quality, and an analysis of how subsampling affects alignment. The project should establish which

anatomical structures can be aligned reliably and where resolution differences or modality-specific contrast limit registration accuracy.

Dataset and resources

- **Dataset:** Oxford Digital Brain Bank (take at least 1 sample of the human corpus callosus)
- **Registration software:** ANTs or SimpleITK or anything that works

Project 5: EEG-to-Image Generation with Generative AI



Objective

Reconstruct an approximate image from EEG recordings of a person viewing natural images, inspired by MindEye's brain-to-image approach but adapted to EEG rather than fMRI.

Proposed methodology

1. Select data: use one THINGS-EEG2 participant and a subset of approximately 500–2,000 training images, with held-out test images.
2. Preprocess EEG: filter, epoch, normalize, and extract spatiotemporal features from the EEG signal.
3. Learn a brain representation: train a lightweight CNN or temporal encoder to predict image embeddings from EEG.
4. Generate images: use the predicted embeddings to condition a pretrained diffusion model or image decoder. Compare semantic reconstruction with lower-level visual reconstruction.
5. Evaluate: compare generated and original images using perceptual similarity, image–text embedding similarity, and qualitative inspection.

Suggested baselines

- A simple EEG-to-image-feature regression model.
- A learned EEG encoder with a frozen pretrained image decoder.
- An optional fine-tuned encoder or diffusion-conditioning model for GPU-equipped teams.

Computational requirements

- The baseline should run with cached pretrained image features and a frozen decoder, without requiring GPU access.
- GPU acceleration is encouraged for training and image generation, but the project should not depend on training a large diffusion model from scratch.
- Use a small data subset and pretrained weights; make model size and data size configurable.

Expected deliverables

- A reproducible notebook for EEG preprocessing, model training, and image generation.
- Side-by-side comparisons of original stimuli and reconstructed images.
- Quantitative and qualitative evaluation.
- A short report on reconstruction quality, generalization, and limitations.

Important: MindEye itself reconstructs images from fMRI, not EEG. The project should use an EEG-specific reconstruction pipeline as its starting point, such as EEG-Image-Reconstruction, rather than attempting to run the original MindEye model unchanged.

Recommended dataset and code



THINGS-EEG2

EEG recordings from 10 participants viewing natural images, with stimulus images and preprocessed recordings available. This is a practical starting point for image reconstruction experiments.

- Dataset and download instructions
- EEG-to-image reconstruction code
- Alternative reconstruction implementation: ENIGMA

The full THINGS-EEG2 dataset is large, so the groups should use a single participant and a selected subset of stimuli. The existing reconstruction repositories provide useful reference implementations, but some use extensive pretrained-model checkpoints and multiple GPUs for full experiments.

Project: Kamil Kuwala test (only one group) Evaluating Brain Tumor Segmentation Models on a New Brain Metastasis Dataset

Project description

Objective. Evaluate the generalizability of an existing brain tumor segmentation model on an independent dataset of brain metastases originating from lung cancer. The project will use the publicly available Brain-Mets-Lung-MRI-Path-Segs dataset from The Cancer Imaging Archive (TCIA).

The dataset contains 111 MRI studies from 103 patients, including contrast-enhanced T1-weighted and FLAIR images with tumor segmentations. The annotations were manually delineated and verified by a board-certified neuroradiologist, making the dataset suitable for evaluating segmentation performance against expert annotations. The scans were acquired across different scanners and magnetic field strengths, providing an opportunity to investigate model generalization. Dataset details.

Tasks

Core task , required for all groups

1. Obtain and explore the brainmets TCIA dataset (NOT THE ENTIRE TCIA), identifying available MRI sequences, segmentation masks, and cases suitable for evaluation.
2. Run the existing segmentation model from Kamil Kuwala's MRI Analyzer repository on the new dataset.
3. Compare the predicted masks with the expert annotations using metrics such as Dice similarity coefficient, Hausdorff distance, and sensitivity.
4. Visualize representative successes and failures, and investigate how performance varies across patients, MRI sequences, and acquisition characteristics.

Extension , for larger groups

If the core evaluation is completed successfully, investigate a lightweight approach to improve performance on the new dataset without training the entire model from scratch. Possible directions include:

- **Pseudo-transfer learning:** adapt feature representations or selected model components to the new dataset.
- **Inference-time adaptation or intensity normalization:** investigate whether simple adjustments improve performance across scanners.
- **Limited fine-tuning:** if feasible, fine-tune selected layers using a small subset of annotated cases and evaluate on held-out patients.
- **Error analysis:** investigate whether segmentation errors are associated with tumor size, lesion location, or image quality.

Any adaptation must be evaluated on patients not used for training or tuning.

Dataset and code

- **Dataset:** Brain-Mets-Lung-MRI-Path-Segs , TCIA
- **Existing model and code:** MRI Analyzer , GitHub

Group organization

This is a **group project**, so please find other students interested in working together.

- **Two students:** focus on testing the existing model on the new dataset, evaluating segmentation performance, and documenting the results. One student can also act as a mentor for a larger group.
- **Three or more students:** complete the evaluation and add an extension, such as pseudo-transfer learning, limited fine-tuning, or systematic error analysis.

The goal is to produce a reproducible evaluation, a clear comparison with expert annotations, and, if time permits, a simple method for improving model generalization to a new clinical dataset.